

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Applications of Differentiation

Differentiation applications turn calculus into gradients, turning points, and optimisation.

APPLICATIONS OF DIFFERENTIATION · P2 · 11

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The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 GRAD: Gradient Function

TRIGGER *rate of change, gradient*

02 STAT: Stationary Point

TRIGGER *turning point, maximum, minimum*

03 OPTIM: Optimisation

TRIGGER *maximum area, minimum cost*

04 SKETCH: Curve Behaviour

TRIGGER *increasing, decreasing, sketch*

BANK EVIDENCE Local pages: 20 problem, 33 answer, 23 notes. Website primary entries: 20.

01 GRAD: Gradient Function

TRIGGER *rate of change, gradient*

BECOMES Differentiate and substitute.

FIRST LINE TO WRITE

$$\text{gradient} = f'(x)$$

SIMPLEST STRATEGY

- 1 Get the derivative.
- 2 Read the x -value.
- 3 Apply substitution.
- 4 Deliver gradient or rate.

WORKED MODEL

| If $y = x^3$, then $dy/dx = 3x^2$.

HARD VARIANTS

- 1 If a point is given, use its x -coordinate in the derivative.
- 2 For rates, keep units such as m s^{-1} .
- 3 If the gradient is given, set $f'(x)$ equal to that value.

— BOTTOM LINE

The derivative is the gradient function.

02 STAT: Stationary Point

TRIGGER *turning point, maximum, minimum*

BECOMES Set derivative to zero.

FIRST LINE TO WRITE

$$f'(x) = 0$$

SIMPLEST STRATEGY

- 1 Set derivative zero.
- 2 Take roots.
- 3 Add y -coordinates.
- 4 Test type.

WORKED MODEL

$$f'(x) = 3x^2 - 12 = 0 \Rightarrow x = \pm 2.$$

HARD VARIANTS

- 1 Use second derivative or sign change to classify.
- 2 Reject roots outside a physical domain.
- 3 For curves with roots or fractions, simplify before differentiating.

— BOTTOM LINE

Stationary points have zero gradient.

03 OPTIM: Optimisation

TRIGGER *maximum area, minimum cost*

BECOMES Write one variable, differentiate, then test.

FIRST LINE TO WRITE

$$\left| \frac{d}{dx}(\text{quantity}) = 0 \right.$$

SIMPLEST STRATEGY

- 1 Object quantity named.
- 2 Put it in one variable.
- 3 Take derivative.
- 4 Identify stationary value.
- 5 Make max/min conclusion.

WORKED MODEL

| Area $A = x(20 - 2x)$, so $A' = 20 - 4x = 0$.

HARD VARIANTS

- 1 Use constraints to remove extra variables first.
- 2 Check endpoints if the domain is closed.
- 3 State the maximum/minimum value in context, not just x .

— BOTTOM LINE

Optimisation starts before calculus: define the quantity.

04 SKETCH: Curve Behaviour

TRIGGER *increasing, decreasing, sketch*

BECOMES Use derivative signs and stationary points.

FIRST LINE TO WRITE

$$f'(x) > 0 \Rightarrow \text{increasing}$$

SIMPLEST STRATEGY

- 1 Solve critical values.
- 2 Keep a sign chart.
- 3 Evaluate derivative sign.
- 4 Trace increasing/decreasing.
- 5 Classify points.
- 6 Highlight intercepts.

WORKED MODEL

| If $f'(x) = (x - 1)(x - 4)$, the sign is positive outside and negative between.

HARD VARIANTS

- 1 Repeated critical roots may not change sign.
- 2 Combine derivative information with intercepts for sketches.
- 3 Use domain restrictions before making a sign chart.

— BOTTOM LINE

The derivative tells the movement of the graph.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
rate of change, gradient	gradient = $f'(x)$
turning point, maximum, minimum	$f'(x) = 0$
maximum area, minimum cost	$\frac{d}{dx}(\text{quantity}) = 0$
increasing, decreasing, sketch	$f'(x) > 0 \Rightarrow$ increasing